

# Marketing Campaign Performance and Budget Recommendations

## February 2021

### 1. Business Objective

Evaluate February 2021 marketing campaign performance across spending efficiency, revenue return, and conversion funnel stages. Use the findings to recommend where to reduce, maintain, or cautiously increase budgets, and identify what must be verified before implementing those changes.

### 2. Data Overview

The dataset was downloaded from Kaggle and covers February 1–28, 2021. It contains 308 records across 11 campaigns and four marketing categories, with each row representing one campaign on one day. Available metrics include advertising spend, impressions, clicks, leads, orders, and revenue.

**Source:** [Digital Marketing Metrics & KPIs to Measure — Kaggle](#)

Monetary amounts are presented in the dataset's original units because the currency has not been verified. "M" denotes millions.

### 3. Data Preparation

Campaign names were standardized using LOWER() and TRIM() to ensure consistent capitalization and remove extra spaces. Dates were converted into Excel date values using DATEVALUE() to support daily analysis.

The dataset was checked for missing values, zero values, invalid inputs, and duplicate campaign-date combinations. Valid zero-order and zero-revenue records were retained. Funnel consistency checks examined whether clicks exceeded impressions, leads exceeded clicks, or orders exceeded leads.

Unusually high impression and spending values were flagged for review and retained because they had not been confirmed as errors. Metrics affected by these values were interpreted cautiously. High impression values directly affect CTR and CPM, while high spending values affect ROAS and cost-based metrics.

## 4. Metric Definitions

Metrics were calculated from aggregated totals at the campaign, category, or date level. Overall ratios were calculated by dividing the relevant totals rather than taking simple averages of campaign or daily ratios.

### Performance Metrics

| Metric                              | Formula                                                         | Interpretation                                  |
|-------------------------------------|-----------------------------------------------------------------|-------------------------------------------------|
| Advertising spend                   | Sum of <code>mark_spent</code>                                  | Total recorded advertising expenditure.         |
| Recorded revenue                    | Sum of <code>revenue</code>                                     | Total revenue recorded in the dataset.          |
| Orders                              | Sum of <code>orders</code>                                      | Total recorded orders.                          |
| Click-through rate (CTR)            | $\text{Clicks} \div \text{impressions} \times 100\%$            | Clicks relative to impressions.                 |
| Cost per click (CPC)                | $\text{Advertising spend} \div \text{clicks}$                   | Average advertising cost per click.             |
| Cost per thousand impressions (CPM) | $\text{Advertising spend} \div \text{impressions} \times 1,000$ | Advertising cost per 1,000 impressions.         |
| Click-to-lead conversion rate       | $\text{Leads} \div \text{clicks} \times 100\%$                  | Recorded leads relative to clicks.              |
| Cost per lead (CPL)                 | $\text{Advertising spend} \div \text{leads}$                    | Average advertising cost per lead.              |
| Lead-to-order conversion rate       | $\text{Orders} \div \text{leads} \times 100\%$                  | Recorded orders relative to leads.              |
| Click-to-order conversion rate      | $\text{Orders} \div \text{clicks} \times 100\%$                 | Recorded orders relative to clicks.             |
| Cost per order (CPO)                | $\text{Advertising spend} \div \text{orders}$                   | Average advertising cost per order.             |
| Average order value (AOV)           | $\text{Recorded revenue} \div \text{orders}$                    | Average recorded revenue per order.             |
| Return on advertising spend (ROAS)  | $\text{Recorded revenue} \div \text{advertising spend}$         | Recorded revenue per unit of advertising spend. |

|                        |                                                                     |                                                                                       |
|------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Spending minus revenue | Advertising spend – recorded revenue                                | Positive values indicate spending exceeded revenue. This is not a profit calculation. |
| Spending share         | Campaign or category spend ÷ total spend for the same period × 100% | Share of the advertising budget allocated to a campaign or category.                  |

### Profitability and Budget Scenario Measures

Actual variable-cost information was unavailable, so contribution margins were treated as assumptions for scenario analysis.

| Measure                                    | Formula                                                           | Interpretation                                                                                    |
|--------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Contribution margin before advertising (%) | (Revenue – variable costs excluding advertising) ÷ revenue × 100% | Percentage of revenue available to cover advertising, fixed costs, and profit.                    |
| Break-even ROAS before fixed costs         | 1 ÷ contribution margin expressed as a decimal                    | ROAS required to cover advertising and assumed variable costs.                                    |
| Minimum required contribution margin       | Advertising spend ÷ revenue × 100%                                | Margin needed to cover advertising at the recorded ROAS.                                          |
| Proposed spending                          | February spending × (1 + proposed change %)                       | Campaign budget under the illustrative scenario.                                                  |
| Budget released or added                   | February spending – proposed spending                             | Positive values indicate released budget; negative values indicate additional allocation.         |
| Unallocated reserve                        | Total budget ceiling – total proposed spending                    | Budget remaining after proposed allocations. A negative result indicates the ceiling is exceeded. |

A minimum required margin above 100% means recorded revenue was below advertising spend even before other costs.

### Daily Benchmark Measures

| Measure                     | Formula                                                 | Interpretation                                                           |
|-----------------------------|---------------------------------------------------------|--------------------------------------------------------------------------|
| Orders at monthly benchmark | Daily leads × the campaign's monthly lead-to-order rate | Hypothetical daily orders at the campaign's own monthly conversion rate. |

|                                    |                                  |                                                                                                |
|------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------|
| Order gap versus monthly benchmark | Benchmark orders – actual orders | Positive values indicate orders below the benchmark; negative values indicate orders above it. |
|------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------|

Benchmark orders were calculated using the full-precision monthly rate. These gaps are diagnostic comparisons, not confirmed lost orders.

ROAS of 1 means recorded revenue equals advertising spend; it does not establish profitability. Funnel rates use aggregate counts and do not confirm that the same individuals progressed through each stage. Ratios with zero denominators are undefined.

## 5. Key Findings

### Overall Performance

Across all 11 campaigns, February advertising spend totaled 30.59M, recorded revenue totaled 42.89M, and recorded orders totaled 8,043.

| Metric              | February result |
|---------------------|-----------------|
| Advertising spend   | 30,590,879.82   |
| Recorded revenue    | 42,889,366.00   |
| Orders              | 8,043           |
| Overall ROAS        | 1.40            |
| Cost per order      | 3,803.42        |
| Average order value | 5,332.51        |

Overall ROAS was 1.40, meaning each unit of advertising spend generated approximately 1.40 in recorded revenue. Although revenue exceeded advertising spend, this does not establish profitability because other variable and fixed costs were not included.

### Campaign Performance Varied Substantially

| Campaign             | Recorded ROAS |
|----------------------|---------------|
| youtube_blogger      | 3.77          |
| facebook_retargeting | 2.01          |
| google_hot           | 1.84          |
| instagram_tier1      | 1.77          |
| instagram_blogger    | 1.37          |
| banner_partner       | 1.22          |
| facebook_tier1       | 0.93          |
| facebook_tier2       | 0.74          |
| google_wide          | 0.66          |
| instagram_tier2      | 0.63          |
| facebook_lal         | 0.11          |

youtube\_blogger had the highest recorded ROAS at 3.77, while facebook\_lal had the lowest at 0.11. Five of the 11 campaigns recorded revenue below advertising spend.

This supports considering a budget reduction for facebook\_lal after verifying tracking accuracy, conversion delays, and campaign objectives. A small increase test for youtube\_blogger could be considered after confirming contribution margins and its capacity to support additional spending. Historical ROAS does not guarantee the same return at a higher budget.

### Social Was the Main Category Requiring Attention

| Category   | Advertising spend | Recorded ROAS | Cost per order | AOV      |
|------------|-------------------|---------------|----------------|----------|
| Social     | 13,798,500.91     | 0.86          | 5,236.62       | 4,520.48 |
| Influencer | 8,305,304.08      | 2.54          | 2,755.58       | 7,007.26 |
| Media      | 5,026,674.76      | 1.22          | 3,209.88       | 3,929.09 |
| Search     | 3,460,400.07      | 1.07          | 4,179.23       | 4,474.72 |

Social had the highest advertising spend but the lowest category ROAS. Its cost per order also exceeded its average order value, consistent with recorded revenue falling below advertising spend.

Four of its six campaigns had ROAS below 1. facebook\_lal and facebook\_tier2 had the largest spending-minus-revenue gaps within social, making them priorities for review.

### **Social's Larger Funnel Gap Occurred at Lead-to-Order**

| <b>Funnel metric</b> | <b>Social</b> | <b>Influencer</b> |
|----------------------|---------------|-------------------|
| Click-to-lead rate   | 2.09%         | 2.26%             |
| Lead-to-order rate   | 8.40%         | 17.79%            |

Social's lead-to-order rate was 9.39 percentage points below influencer's, while its click-to-lead rate was only 0.17 percentage points lower.

This suggests prioritizing investigation of lead quality, purchase barriers, and conversion timing. However, the aggregate data cannot establish the cause of the difference.

### **instagram\_tier2 Combined High Lead Volume with Weak Order Conversion**

instagram\_tier2 generated 10,374 leads—approximately 33% of social's total—but only 313 orders, approximately 12% of social's orders. Its lead-to-order rate was 3.02%, compared with 8.40% for social overall.

Despite having the lowest campaign CPC at 2.09, its ROAS was only 0.63. Inexpensive clicks and high lead volume therefore did not translate into strong recorded revenue returns.

Daily analysis identified February 5, 15, and 18 as dates with substantial lead volume but orders below the campaign's monthly conversion benchmark.

| <b>Date</b>     | <b>Leads</b> | <b>Actual orders</b> | <b>Benchmark orders</b> | <b>Benchmark minus actual orders</b> |
|-----------------|--------------|----------------------|-------------------------|--------------------------------------|
| February 5      | 765          | 4                    | 23.1                    | 19.1                                 |
| February 15     | 916          | 11                   | 27.6                    | 16.6                                 |
| February 18     | 1,020        | 16                   | 30.8                    | 14.8                                 |
| <b>Combined</b> | <b>2,701</b> | <b>31</b>            | <b>81.5</b>             | <b>50.5</b>                          |

This difference is a diagnostic signal, not a confirmed estimate of lost orders. Before attributing it to lead quality or purchase barriers, the team should verify conversion delays and tracking accuracy using linked lead and order records.

### **Daily ROAS Reflected Both Category Returns and Spending Mix**

Overall daily ROAS ranged from 1.11 on February 10 to 1.95 on February 26.

| <b>Measure</b>                  | <b>February 10</b> | <b>February 26</b> |
|---------------------------------|--------------------|--------------------|
| Total advertising spend         | 729,951.80         | 122,692.89         |
| Overall ROAS                    | 1.11               | 1.95               |
| Influencer ROAS                 | 1.87               | 2.81               |
| Influencer share of daily spend | 20.10%             | 44.34%             |

Both stronger influencer returns and the increased share of spending allocated to influencer contributed to the higher overall ROAS on February 26.

However, total spending was substantially lower on that date, and other categories' results also changed. This comparison does not establish that reallocating budget to influencer would reproduce the same improvement.

## **6. Budget Recommendations**

### **facebook\_lal: Test a Budget Reduction**

facebook\_lal spent 2.64M but generated only 0.30M in recorded revenue, resulting in a ROAS of 0.11.

Before reducing spending, verify revenue attribution, conversion delays, and campaign objectives. If these checks confirm the reported underperformance, an initial 10% reduction would release approximately 264,193.92.

This percentage is an illustrative starting point, not an optimized budget cut.

### **youtube\_blogger: Consider a Controlled Increase Test**

youtube\_blogger had the highest recorded ROAS at 3.77. The illustrative scenario increases its February budget by 5%, adding 202,896.83.

Before implementing this increase, verify attribution, contribution margins, and capacity to support additional spending. Its historical ROAS requires a contribution margin before advertising of approximately 26.50% to cover advertising costs before fixed costs.

Monitor the additional spending against the break-even ROAS implied by the verified margin. Historical performance does not guarantee the same return as spending increases.

### **Other Campaigns: Continue Review Before Reallocation**

facebook\_tier1, facebook\_tier2, google\_wide, and instagram\_tier2 also recorded ROAS below 1 and warrant review for potential reductions. Their budgets remain unchanged in the illustrative scenario pending further verification.

Campaigns with ROAS above 1 should not automatically receive increases. Actual margins, attribution, campaign objectives, and performance under additional spending must support any expansion.

### **Illustrative Allocation Scenario**

| <b>Budget item</b>                       | <b>Amount or change</b> |
|------------------------------------------|-------------------------|
| Original total budget                    | 30,590,879.82           |
| facebook_lal budget change               | -10%                    |
| Budget released from facebook_lal        | 264,193.92              |
| youtube_blogger budget change            | +5%                     |
| Additional allocation to youtube_blogger | 202,896.83              |
| Total proposed campaign spending         | 30,529,582.72           |
| Unallocated reserve                      | 61,297.10               |

All other campaign budgets remain unchanged. Calculations use full precision; displayed figures are rounded to two decimal places.

The reserve remains unallocated rather than assuming all released funds must immediately be spent. The scenario does not forecast revenue or demonstrate an optimized allocation.

## **7. Limitations**

- **Limited period:** The dataset covers only February 2021. Results may not generalize to other months or seasonal conditions.

- **Source context:** The dataset was obtained from Kaggle. Its original collection process, currency, and business context have not been independently verified.
- **Profitability is unknown:** Variable costs and fixed costs were unavailable. Contribution margins were hypothetical scenario inputs.
- **Aggregate funnel data:** Daily leads and orders are not linked to individual customers. Conversion delays may affect observed daily rates.
- **Unverified unusual values:** Exceptionally high impression and spending values were retained because they were not confirmed as errors.
- **Attribution uncertainty:** Recorded revenue does not establish how much revenue advertising caused.
- **Illustrative recommendations:** Budget changes were proposed, not implemented or experimentally tested. They do not represent proven improvements or an optimized allocation.

## 8. Conclusion

February's campaigns generated 42.89M in recorded revenue from 30.59M in advertising spend, producing an overall ROAS of 1.40. Performance varied substantially: youtube\_blogger achieved the highest ROAS, while facebook\_lal had the lowest. Social combined the largest spending allocation with the weakest category ROAS, warranting further investigation of campaign performance and lead-to-order conversion.

The proposed budget scenario pairs a cautious reduction in facebook\_lal with a small increase test for youtube\_blogger, retaining an unallocated reserve. Implementation should depend on verified tracking, conversion timing, contribution margins, and campaign objectives. These recommendations support controlled testing; they do not establish profitability or guarantee future improvements.